

Lec 11. Introduction to Neural Networks

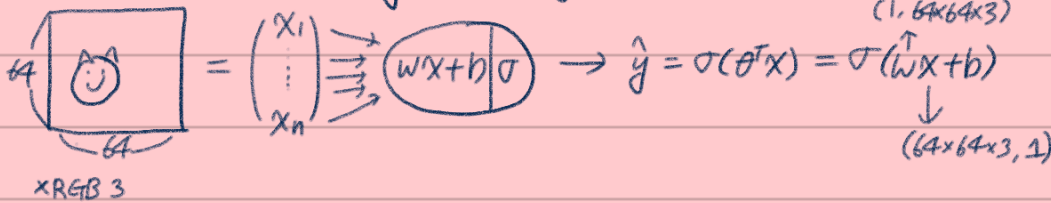
Outline I. Logistic Regression

II. Neural Networks

Deep Learning { Computer vision
NLP
speech recognition

Logistic Regression

goal 1.0: find cats in images (binary classification: 0 → absence of a cat, 1 → presence of the cat)



i) initialize W (weight), b (bias)

ii) Find the optimal w, b → $L = -[y \log \hat{y} + (1-y) \log(1-\hat{y})]$ *minimize* from MLE

iii) Use $\hat{y} = \sigma(Wx + b)$ to predict

by Gradient Descent $\begin{cases} W = W - \alpha \frac{\partial L}{\partial W} \\ b = b - \alpha \frac{\partial L}{\partial b} \end{cases} \rightarrow \sigma' = \sigma(1-\sigma)$

(Eq 1) neuron = linear + activation

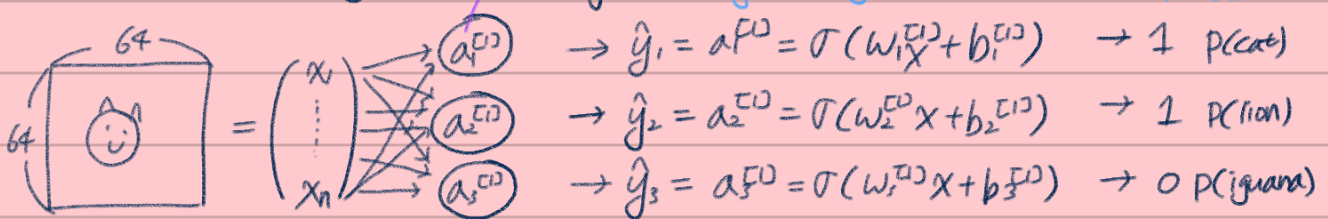
(Eq 2) model = architecture + parameters

In a for loop,

you will compute the derivative of the loss with respect to your parameters.

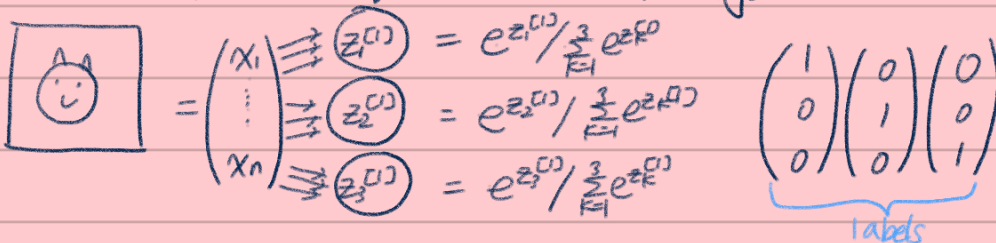
goal 2.0: find cat/lion/iguana in images (3 logistic regression) *'activation'*

labels



$$L_{3N} = -\sum_{k=1}^3 [y_k \log \hat{y}_k + (1-y_k) \log(1-\hat{y}_k)] \text{ when the models are independent}$$

goal 3.0: + constraint unique animal on an image softmax multi-class regression



cross-entropy loss

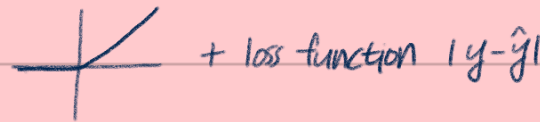
$$L_{CE} = -\sum_{k=1}^3 y_k \log \hat{y}_k$$

Can't predict cat's age $\rightarrow$ ① softmax multiclass regression

$\therefore$ age is unique! we want the probability distribution along the edge (ages)

② use regression directly

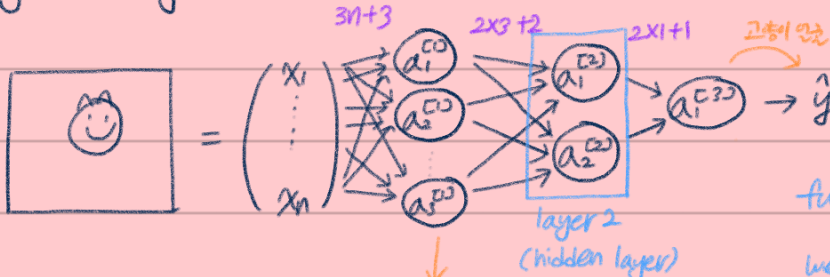
ReLU (Rectified Linear Units) $\rightarrow$ linear regression on the edge of 0 or 1



deep learning on the other side of the coin

Neural Networks "black box models" "end-to-end learning"

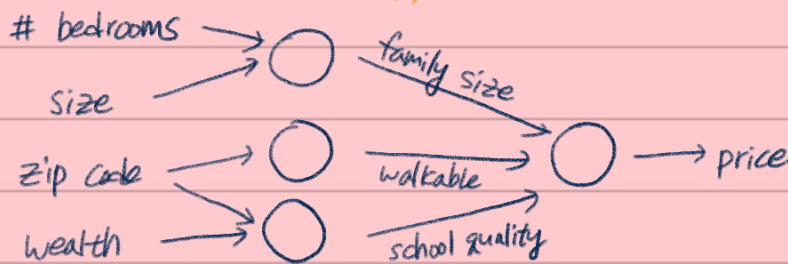
goal image $\Rightarrow$ cat (1) vs. no cat (0)



fully connected layer

We will let the neurons of the first layer figure out what's interesting for the second layer to make the prediction

House price prediction



Propagation equations

$$z^{[1]} = w^{[1]}x + b^{[1]} \quad z^{[1]} \text{ 은 } (3,1)$$

$$a^{[1]} = \sigma(z^{[1]}) \quad a^{[1]} \text{ doesn't change the shape } (\because \text{just sigmoid}) (3,1)$$

$$z^{[2]} = w^{[2]}a^{[1]} + b^{[2]} \quad (2,1) \text{ 로부터 } w^{[2]} \text{ 을 구함}$$

$$a^{[2]} = \sigma(z^{[2]}) \quad (2,1)$$

$$z^{[3]} = w^{[3]}a^{[2]} + b^{[3]} \quad (1,1)$$

$$a^{[3]} = \sigma(z^{[3]})$$

Stochastic gradient descent is updates the weights and the bias after you see every examples so the direction of the gradient is quite noisy.

vs.

(batch) gradient descent updates after you've seen the whole batch of examples so the gradient is much more precise

what happens for an input batch of m examples?

$$X = \begin{pmatrix} | & | & & | \\ x^{(1)} & x^{(2)} & \dots & x^{(m)} \\ | & | & & | \end{pmatrix}$$

id of example (pointing to the column index)

square brackets - layer

round brackets (super script) - id of example

↑
input image

$$z^{[1]} = \underset{(3,m)}{w^{[1]}} \underset{(3,n)}{x} + \underset{(3,1)}{b^{[1]}} \quad \text{Broadcasting} \quad \tilde{b}^{[1]} = \begin{pmatrix} | & \dots & | \\ b^{[1]} & \dots & b^{[1]} \\ | & \dots & | \end{pmatrix}$$

$\underbrace{\hspace{10em}}_m$

$$= \begin{bmatrix} | & & | \\ z^{1} & \dots & z^{[1](m)} \\ | & & | \end{bmatrix}$$

Optimizing $w^{[1]}, w^{[2]}, w^{[3]}, b^{[1]}, b^{[2]}, b^{[3]}$

Define loss/cost function

$$J(\hat{y}, y) = \frac{1}{m} \sum_{i=1}^m \mathcal{L}^{(i)} \quad \text{with } \mathcal{L}^{(i)} = -[y^{(i)} \log \hat{y}^{(i)} + (1 - y^{(i)}) \log (1 - \hat{y}^{(i)})]$$

$$\frac{\partial J}{\partial w^{[2]}} \text{ vs. } \frac{\partial \mathcal{L}}{\partial w^{[2]}}$$

Backward Propagation

$$\forall l = 1 \dots 3 \quad \begin{cases} w^{[l]} = w^{[l]} - \alpha \cdot \frac{\partial J}{\partial w^{[l]}} \\ b^{[l]} = b^{[l]} - \alpha \cdot \frac{\partial J}{\partial b^{[l]}} \end{cases}$$

$$\frac{\partial J}{\partial w^{[3]}} \quad \begin{aligned} &= \frac{\partial J}{\partial ?} \cdot \frac{\partial ?}{\partial w^{[3]}} = \frac{\partial J}{\partial a^{[3]}} \cdot \frac{\partial a^{[3]}}{\partial z^{[3]}} \cdot \frac{\partial z^{[3]}}{\partial w^{[3]}} \\ &\quad \text{chain rule} \\ &= \frac{\partial J}{\partial z^{[3]}} \cdot \frac{\partial z^{[3]}}{\partial a^{[2]}} \cdot \frac{\partial a^{[2]}}{\partial z^{[2]}} \cdot \frac{\partial z^{[2]}}{\partial w^{[2]}} \end{aligned}$$

$$\frac{\partial J}{\partial w^{[1]}} = \frac{\partial J}{\partial z^{[2]}} \cdot \frac{\partial z^{[2]}}{\partial a^{[1]}} \cdot \frac{\partial a^{[1]}}{\partial z^{[1]}} \cdot \frac{\partial z^{[1]}}{\partial w^{[1]}}$$